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# Mass-spectrometry-based proteomics enables rapid and accurate diagnosis of Lyme neuroborreliosis in adults

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# Abstract

Lyme neuroborreliosis (LNB), a severe nervous system infection caused by tick-borne spirochetes of the *Borrelia burgdorferi* sensu lato complex, represents one of the most frequent bacterial infections of the nervous system in Europe. Early diagnosis remains challenging due to limited sensitivity of current methods and requires invasive lumbar punctures, underscoring the need for improved, less invasive diagnostic tools.

Here, we applied mass spectrometry-based proteomics to analyse 308 cerebrospinal fluid (CSF) samples and 207 plasma samples from patients with LNB, viral meningitis, controls and other manifestations of Lyme borreliosis. Diagnostic panels of regulated proteins were identified and evaluated through machine learning-assisted proteome analyses.

In CSF, the classifier distinguished LNB from viral meningitis and controls with AUCs of 0.92 and 0.90, respectively. In plasma, LNB was distinguished from controls with an AUC of 0.80.

Our findings highlight the diagnostic potential of machine learning-assisted proteomics for LNB in CSF and plasma.


# INTRODUCTION

Lyme neuroborreliosis (LNB) is a bacterial infection of the nervous system caused by spirochetes of the *Borrelia burgdorferi* sensu lato (*B. burgdorferi* s.l.) complex transmitted through bites from hard-shelled ticks of the *Ixodes* genus[1]. With an approximate incidence in endemic countries between 3.2 and 6.3 per 100 000 persons/year[2–4] LNB is among the most frequent bacterial infections of the nervous system in Europe[5,6]. LNB can cause a wide range of clinical neurological conditions, but most frequently presents as a subacute painful meningo-radiculitis with radiating pain from the spine to neck, extremities, thorax or abdomen, lymphocytic meningitis and/or cranial neuropathies e.g., facial nerve palsy[7]. If relevant antibiotic therapy is administered at an early stage of disease, LNB has a favourable long-term prognosis[8–10]. However, delayed treatment is associated with an increased risk of residual symptoms and long-term sequelae[9]. In countries where LNB is endemic, the average time from onset of neurological symptoms to diagnosis is typically around 3 weeks and has remained unchanged for the last four decades[6].

The cause of the diagnostic delay is multifactorial. Overlapping symptomatology with other more common diseases and the fact that only around 40% of patients with LNB report a tick bite and only 25% report a history of the classic skin rash erythema migrans makes it less likely that physicians and patients consider LNB as a differential diagnosis, especially in the absence of facial nerve palsy[6,8].

However, even when LNB is clinically suspected at an early stage, the diagnosis requires detection of *B. burgdorferi* s.l.- specific antibodies in the cerebrospinal fluid (CSF). Lumbar puncture is an

uncomfortable and expensive procedure that may require hospitalization and generel anesthesia for children. Direct pathogen identification of *B. burgdorferi* s.l. with polymerase chain reaction (PCR), cultivation or a combination has retained an exceedingly poor level of sensitivity both in blood and CSF despite several attempts to improve tools and techniques over the years[11–14]. Further, the serological response in LNB is not detectable early in the course of disease and antibodies may remain elevated for months to several years after full recovery of well-treated infections making it impossible to discriminate between past and current infection. Furthermore, the *B. burgdorferi* s.l.-specific IgG and IgM in blood, poorly predict nervous system involvement[15].

Thus, in order to reduce the diagnostic delay in LNB, the need for novel diagnostic tools is evident. Especially improved blood-based diagnostic tests would be extremely valuable as they would both potentially help reduce the diagnostic delay and offer a less invasive diagnostic tool.

Untargeted proteomic analysis enables accurate measurement of proteins in a sample and can be used to compare patterns of protein regulation in patients with and without disease to identify disease-specific protein response signatures[16]. When combined with machine-learning (ML) for data-evaluation and analysis, proteomic test results can be made available with turnaround times of just 2 hours making it highly attractive in real-life clinical settings.

The aim of this study was to explore the potential of ML-assisted MS-based proteomics as a novel diagnostic tool for LNB in CSF and plasma of adults.

# RESULTS

Study population

A total of 483 CSF and plasma samples from adult individuals were eligible for proteomics analysis: 155 samples from patients with LNB, 127 samples from patients with viral meningitis, 169 controls and 32 samples from patients with erythema migrans, acrodermatitis chronica atrophicans and post-treatment-Lyme-disease-syndrome (Table 1).

**Table 1** Baseline characteristics of patients with Lyme neuroborreliosis, viral meningitis, controls, and other Lyme borreliosis manifestations.

| Cohort | Diagnosis | n | Age Mean [IQR] | Females n (%) |
|---|---|---|---|---|
| **CSF Development** | Lyme neuroborreliosis | 49 | 51[40-66] | 26 (53) |
| **N = 145** | Viral meningitis | 44 | 38[27-42] | 21 (47) |
| | Control | 52 | 36[25-42] | 17 (32) |
| **CSF Validation** | Lyme neuroborreliosis | 69 | 53[46-67] | 28 (40) |
| **N = 163** | Viral meningitis | 45 | 31[25-37] | 23 (51) |
| | Control | 49 | 44[33-53] | 24 (48) |
| **Plasma Development** | Lyme neuroborreliosis | 27 | 64[61-69] | 10 (37) |
| **N = 95** | Viral meningitis | 20 | 35[27-38] | 13 (65) |
| | Control | 48 | 46[33-60] | 28 (58) |
| **Plasma Validation** | Lyme neuroborreliosis | 10 | 59[56-63] | 4 (40) |
| **N = 80** | Viral meningitis | 18 | 38[24-43] | 10 (55) |
| | Control | 20 | 36[28-42] | 6 (30) |
| | Acrodermatitis chronica atrophicans | 6 | 66[59-74] | 6 (100) |
| | Erythema migrans | 9 | 47[36-54] | 5 (55) |
| | Post-treatment-Lyme-disease-syndrome | 17 | 58[48-64] | 10 (58) |

Abbreviations: CSF = cerebrospinal fluid, IQR = Inner Quartile Range

## CSF proteomics

*CSF development cohort*

A total of 1,865 proteins were identified in CSF, with a median of 773 quantified pr. sample (Supplementary Figure 1). Fourteen samples were excluded due to low protein numbers. After filtering for missingness, a total of 654 proteins were included for further analysis.

A total of 176 proteins were significantly different between LNB and viral meningitis, of these 10 had an absolute log2 fold change larger than 1 (Figure 1A, Supplementary Table1). A total of 464 proteins were significantly different between LNB and controls, of these 41 had an absolute log2 fold change larger than 1 (Figure 1B, Supplementary Table1).

Of the significant proteins with high log2 fold changes in the two comparisons, 7 proteins overlapped (Figure 1C).

*CSF validation cohort*

In the validation cohort a total of 147 proteins were significantly different in LNB compared to viral meningitis and controls with high log2 fold changes, where 43/46 proteins from the development cohort overlapped (Figure 1B and Supplementary Figure 2,3).

*Protein signatures of LNB in CSF*

Protein signatures were dominated by immunoglobulins (e.g., IGLV3-25, IGLV2-18, FCGBP, IGHM), proteins involved in innate immune responses (e.g. enolase 1, S100A9), neuroendocrine signalling (e.g. ECRG4, CHGA), cell-migration and cell-damage (e.g. PFN1, APCS, YWHAZ, H4C1, ACTA2) (Figure 1C). Immunoglobulins were generally upregulated, while proteins involved in innate immune responses and neuroendocrine signalling were relatively downregulated in LNB compared to viral meningitis and controls.

## Diagnostic classifier for LNB based on machine learning and CSF proteomics

The protein profiles of the CSF development cohort were used to develop a ML model to evaluate the potential of MS-based proteomics coupled with ML as diagnostic support in LNB. Of the twelve ML models tested, the Logistic Regression (LR) model obtained the highest performance in the classification of viral meningitis vs. LNB, whereas the Support Vector Classifier (SVC) model produced the best result in the control vs. LNB classification (Supplementary Figure 4 and 5 respectively). The performance of the diagnostic classification model of viral meningitis vs. LNB on the test set had an area under the curve (AUC) of 0.91 (std. = 0.11) and a Matthews Correlation Coefficient (MCC) of 0.81 (std.=0.08) (Figure 2A). When applied on the validation cohort, the model obtained an AUC and MCC of 0.92 (std. = 0.02) and 0.7 (std.=0.06). According to their SHAP (SHapley Additive exPlanations) values, the most important proteins in the discrimination between viral meningitis and LNB are visualized in Figure 2B. The performance of the diagnostic classification model of control vs. LNB on the test set had an AUC of 0.93 (std. = 0.07) and a MCC of 0.76 (std.=0.14) (Figure 2C). When applied on the validation cohort, the model obtained an AUC and MCC of 0.9 (std. = 0.01) and 0.63 (std.=0.06). The most important proteins in the model according to their SHAP values (Figure 2D) were similarly related to innate and humoral immune responses, neuroendocrine signalling, and cell-damage.

## Plasma proteomics

*Plasma development cohort*

A total of 379 proteins were identified in plasma, with a median of 268 proteins quantified per sample (Supplementary Figure 6). One sample was excluded due to low protein number. After filtering for missingness, a total of 232 proteins were included for further analysis.

A total of 61 proteins were significantly different between LNB and viral meningitis (Figure 3A, Supplementary Table2) and 63 for LNB versus controls (Figure 3B). The overlap between these two

comparisons of significant proteins were 36 (Figure 3C, Supplementary Table2).

*Plasma validation cohort*

In the validation cohort a total of 25 proteins were significantly different in LNB compared to viral meningitis and controls. Ten overlapped with the significant proteins from the development cohort (Figure 3B and Supplementary Figure 7,8).

*Protein signatures of LNB in plasma*

The differences in protein profiles between the three patient groups are dominated by proteins involved in innate and humoral immune responses (Figure 3C). When comparing protein profiles of plasma samples from LNB and viral meningitis, patients with LNB generally seem to have a relative upregulation of proteins associated with innate immunity and complement activation (e.g., FCN3, SERPING1, SERPINA5), lipid metabolism (e.g., APOE, APOC1, APOM), and coagulation regulation (e.g., F13A1, PROC). In contrast, patients with viral meningitis have a plasma protein profile dominated by acute-phase and pro-inflammatory markers (e.g., CRP, S100A8, S100A9) and immunoglobulin production. In the comparison between LNB and controls, upregulated plasma-proteins in LNB are similarly involved in complement activation (e.g., C3, C5), coagulation and inflammation regulation (e.g., SERPINA3, SERPIND1), and immune system modulators (e.g., HP, ITIH4). In contrast to CSF, most immunoglobulins had a higher protein level in the plasma of control individuals compared to LNB patients.

## Diagnostic classifier for LNB based on machine learning and plasma proteomics

Similar to the CSF analysis, plasma proteome data were subjected to ML algorithm- and hyperparameter selection in a 3-fold cross validation scheme. It was not deemed feasible to develop a model for the discrimination of viral meningitis and LNB due to the low number of viral meningitis cases, hence, the only model tested was the discrimination of controls and LNB. Of the twelve models tested, the SVC model obtained the best performance in the test set (Supplementary Figure 9) with an AUC of 0.96 (std.=0.03) and a MCC of 0.83 (std.=0.008) (Figure 4A). When applied on the validation cohort the model obtained an AUC of 0.80 (std.=0.02) and a MCC of 0.48 (std.=0.04) (Figure 4A). According to the SHAP feature importance values, the most predictive proteins for the diagnostic classification of controls vs. LNB (Figure 4B) were associated with innate immunity, humoral immune defence, coagulation and cellular metabolism. When the LNB classification model was applied to other diagnostic groups, it identified groups with active *Borrelia burgdorferi* s.l. infections; acrodermatitis chronica atrophicans and erythema migrans, as having significantly higher similarities to the LNB profile than groups with absence of active infection; Post-treatment-Lyme-disease-syndrome or unrelated aetiology; viral meningitis. Mann-Whitney U Test (acrodermatitis chronica atrophicans+erythema migrans vs Post-treatment-Lyme-disease-syndrome +Viral meningitis): U statistic = 5475.0, p-value = 2.65e-07. (Figure 4c).

## Comparison of protein profiles in CSF and plasma

We found an overlap of 72 proteins that were significantly different between LNB and controls or viral meningitis in both CSF and plasma (Figure 5A, Supplementary Table 3). The majority of overlapping proteins (62/72) were relatively upregulated in CSF from patients with LNB compared to controls and viral meningitis, and 30 of these were inversely regulated in plasma. This group primarily included immunoglobulins (18/30) (Figure 5B).

# DISCUSSION

Our study shows that MS-based proteomics can accurately differentiate patients with LNB from patients with viral meningitis and non-LNB controls. Notably, we observed a high level of diagnostic accuracy in plasma, highlighting the potential of blood-based proteomics as a clinically valuable tool. Given the current need for novel diagnostic approaches to reduce delays and reliance on lumbar punctures, these findings could have significant implications for both initial diagnosis and monitoring of treatment response in LNB.

To our knowledge, this is the first study reporting the use of untargeted proteomics as a diagnostic tool in a large and well-defined cohort of adult LNB patients, investigating both CSF and plasma samples. Additionally, the inclusion of viral meningitis cases strengthens our findings by assessing the ability of proteomics to distinguish between two conditions with overlapping symptomatology and CSF findings. This approach enhances the translational potential of our results.

Previous studies investigating the diagnostic potential of proteomics in LNB have been limited in scope and methodology. A small US-based study assessing the diagnostic utility of proteomics in Lyme disease including a small sub-group with CNS involvement; *Angel et al.* (2013) found a panel of 13 proteins discriminated between cases and controls reaching an AUC of 0.8[17]. However, the focus on other Lyme disease manifestations than LNB likely reflects the, somewhat debated, geographical difference in prevalence of neurological involvement with a *B. burgdorferi* s.l.strain-dependent lower incidence in North America compared to Europe[18,19]. More recent European studies have investigated the diagnostic use of protein panels in LNB, though with different methodologies and sample types: *Fredriksson et al.* (Sweden, 2024) analysed serum from 119 paediatric patients with LNB (n=61) and non-LNB (n=58) using a multiplex proximity extension assay that included a panel of 92 predefined inflammatory markers and found that a 5-protein-panel identified LNB with an AUC of 0.88[20]. Though the diagnostic groups and the use of blood-samples were comparable to our study, it is unclear to what extent results in children can be extrapolated to adults and the multiplex assay used is not comparable to the untargeted MS-based proteomics used in our study. *Gęgotek et al.* (Poland, 2024) performed LC-MS based proteomics on serum from patients with LNB (n=10) and controls (n=27) and found significant differences, but did not report measures of diagnostic accuracy[21]. Other investigations have primarily focused on characterizing patients with post-treatment-Lyme-disease-syndrome rather than acute LNB[22,23]. Despite differences in study design, a consistent pattern of immune activation has emerged

across studies, emphasizing the role of innate immune responses, complement system and humoral immune defence in the human host response to *B. burgdorferi* s.l. infection.

Our proteomic findings in CSF, indicate a distinct immune response characterized by elevated levels of immunoglobulin chains, complement-related proteins, and proteins involved in immune cell migration and cytoskeletal dynamics. These findings suggest a highly targeted immune response against *B. burgdorferi* s.l. antigens, involving adaptive immunity with significant antibody production and complement activation. This aligns with established knowledge of host immune response during infections with *B. burgdorferi* s.l.[24]. Interestingly, the antibody response exhibited opposite regulation in CSF and plasma, likely illustrating the significant compartmentalization of the humoral immune response to the CNS in LNB and reflects the fact that all patients with LNB in this study had a positive *B. burgdorferi* s.l.- specific intrathecal antibody index. Moreover, we observed overlapping regulation of key immune-related proteins in both plasma and CSF, suggesting a coordinated immune response across compartments. Plasma and CSF both showed changes in immunoglobulin chains, complement factors, and innate immune modulators in LNB compared to controls, indicating a systemic and localized immune activation. The CSF findings highlight immunoglobulin diversity and a direct antibody response, reflecting the local immune reaction in the CNS. Meanwhile, plasma proteomics revealed a strong systemic immune engagement, with upregulation of innate immune components such as ficolins and coagulation proteins, underscoring systemic inflammation. Notably, viral meningitis exhibited a different pattern, with both plasma and CSF showing elevated acute-phase reactants (e.g., CRP, S100A9) and immunoglobulins, consistent with a strong inflammatory and humoral response. Opposing regulation of certain proteins in CSF and plasma in LNB likely reflects a compartmentalized immune strategy, where the CNS immune response is tightly regulated, while plasma changes mirror systemic inflammation. Additionally, factors such as blood-brain barrier permeability, local versus systemic protein production, and clearance mechanisms may further influence these patterns.

Additionally, structural proteins involved in immune cell motility were upregulated in CSF, indicating active immune cell recruitment within the CNS, specifically in LNB. Viral meningitis patients present a pattern more focused on inflammatory markers and metabolic activity, consistent with an acute viral immune response.

A small cluster of cytoskeleton and extracellular matrix proteins (VIM, ACTA2, PFN1, FN1 and ACTB) emerged among the top discriminatory markers, suggesting increased cell-turnover in LNB. While expected in LNB vs. controls, this is more surprising when compared to viral meningitis. Notably, *B. burgdorferi* s.l. spirochetes are hypothesized to extravasate early upon transmission and travel through the extracellular matrix to peripheral nerves eventually reaching the borders of the CNS and creating the inflammatory basis of radiculoneuritis[18,25]. Vimentin (VIM), a class III intermediate filament that regulates myelination in axons of peripheral nerves[26], is particularly relevant given our limited understanding of CNS entry by *B. burgdorferi* s.l. spirochetes, the frequent cranial nerve involvement in LNB with painful radiculoneuritis[9,27–29] and because proteins released from damaged nerves to the peripheral circulation could be potential biomarker candidates for early LNB, as previously shown with

neurofilament light chain (NfL)[30]. Additionally, fibronectin (FN1) which is also found in this cluster, is considered essential for *B. burgdorferi* s.l. cell adhesion and migration through binding to *B. burgdorferi* s.l. surface proteins BBK32[31] and RevA[32].

The clinical implications of our findings are considerable. If validated in independent cohorts, CSF proteomics could aid in diagnosing LNB among adult patients with CNS infections with unknown aetiology. This may be particularly relevant for PCR-negative suspected viral meningitis cases, where CSF proteomics suggestive of LNB could prompt further testing or the direct initiation of relevant antibacterial treatment e.g. doxycycline. However, the most significant clinical implication would be the ability to diagnose LNB using a blood sample alone - a true gamechanger in LNB diagnostics. It would spare patients the discomfort and risk for complications of lumbar puncture and reduce healthcare-costs associated with referral to specialized healthcare facilities. A diagnostic blood test for LNB would likely reduce the diagnostic delay and thereby reduce the risk of residual symptoms. Even a modest improvement over current serology-based approaches, could lead to fewer invasive procedures and significantly ease treatment-effect monitoring. This would be particularly beneficial in children, who account for around 30% of all LNB patients, and often require general anaesthesia to have a lumbar puncture performed[33 34,35]. Moreover, a blood-based test could address the unmet need for a less invasive method to monitor treatment response in patients with persistent symptoms after LNB treatment.

While these findings demonstrate the diagnostic potential of proteomics in LNB, it is important to consider the strengths and limitations of our study. With 483 samples from patients with LNB, viral meningitis and non-LNB controls, our study represents the largest investigation to date. While we could not stratify patients in full accordance with the European Federation of Neurological Societies (EFNS) diagnostic criteria[7] due to limited clinical data, our use of uniform case definition and the specificity of a first time positive *B. burgdorferi* sl. intrathecal antibody index minimizes the risk of false-positive LNB diagnoses[36]. Our selection of clinically relevant comparison groups, including viral meningitis and patients who were clinically suspected for LNB, enhances the real-world applicability of our findings by testing the robustness of proteomics in distinguishing LNB from other CNS infections. A key strength of our study is its focus on diagnostic feasibility, employing a proteomic workflow with a short turnaround time suitable for clinical implementation. However, the limited detection range inherent to this approach means that certain low-abundance but highly disease-specific proteins, such as CXCL13, may not be captured. Given that previous proteomic studies in LNB have also failed to detect CXCL13, this highlights a general challenge in proteomic biomarker discovery rather than a limitation of our study specifically[37]. As with all retrospective analyses, our study is constrained by sample availability and cohort composition. Differences in cohort sizes necessitated varying cross-validation strategies, and some observed differences in protein signatures may be influenced by factors such as disease stage, age, and sex[38–42]. However, our study population reflects real-world clinical variation, as patients were primarily recruited from referral hospitals. Furthermore, prioritizing clinically relevant diagnostic groups over strictly age- and sex-matched controls enhances the translational value of our findings.

In conclusion, our findings demonstrate that MS-based proteomics can accurately distinguish LNB from controls in both CSF and plasma. These results warrant further prospective validation, particularly with longitudinal sampling, to assess the diagnostic utility of plasma proteomics and its potential role in monitoring treatment response in LNB patients.

# Declarations


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### Data and code availability

The raw data and processed data have been deposited to the ProteomeXchange Consortium via the Proteomics Identifications Database (PRIDE) partner repository, the dataset identifier will be added upon publication[54]. The Jupyter notebooks are available at github upon request.

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# MATERIALS AND METHODS

### Ethics

Samples were obtained during diagnostic investigation and all patients had given their informed consent to the storage of biological material and its future use in research. The study was approved by Knowledge Center for Data Reviews (P-2019-707) and the local ethics committee (H-17024315). The biobanks were approved by the Danish Data Protection Agency (Rigshospitalet: j.nr.: 2012-41-0036).

### Study design

This study was an observational retrospective cohort study.

### Study population and setting

We identified a total of 483 CSF (n = 308) and plasma (n = 175) samples from adults (>18 years) with LNB, erythema migrans, acrodermatitis chronica atrophicans, post-treatment-Lyme-disease-syndrome, viral meningitis, and individuals who were investigated for suspected LNB, but had normal CSF (controls) from; i) The Danish National Biobank (samples collected from 2001-2011)[43] and ii) the Biobank of the Department of Infectious Diseases at Copenhagen University Hospital, Rigshospitalet (samples collected from 2016-2022). After informed patient consent for storage and future use in research was provided, CSF samples were labelled and stored at -80°C and blood samples in EDTA tubes were spun and the supernatant was transferred to Eppendorf tubes, labelled, and stored at -80°C. Samples from the two sites of origin were kept separate in the subsequent analyses to preserve independent cohorts for model development and validation.

### Cohorts

Samples were divided in four cohorts for diagnostic model development and validation in CSF and plasma respectively. The CSF development cohort consisted of 145 CSF samples from patients diagnosed with LNB (n=49), viral meningitis (n=44) and controls (n=52). The CSF validation cohort included a total of 163 CSF samples from patients diagnosed with LNB (n=69), viral meningitis (n=45), and controls (n=49). The plasma development cohort included a total of 95 plasma samples from patients with LNB (n=27), viral meningitis (n=20) and controls (n=48) whereas the plasma validation cohort consisted of 80 plasma samples from patients with LNB (n=10), viral meningitis (n=18), controls (n=20) and an additional cohort of plasma samples from patients with manifestations of Lyme borreliosis without CNS involvement including post-treatment-Lyme-disease-syndrome (n=17), erythema migrans (n=9), and acrodermatitis chronica atrophicans (n=6).

### Definitions of diagnostic groups

**LNB:** Patients diagnosed with LNB based on a first-time positive *B. burgdorferi*-specific intrathecal antibody test and the International Classification of Diseases 10$^{th}$ revision (ICD-10) diagnosis code Borreliosis: A69.

**Viral Meningitis:** Patients with a final clinical diagnosis of viral meningitis determined by an infectious disease specialist at a tertiary referral university hospital. The diagnosis was based on clinical presentation, CSF pleocytosis and exclusion of other diagnoses.

In 76 cases a PCR-verified viral aetiology was determined (Herpes simplex virus 2 (n=32), Enterovirus (n=30), Varicella zoster (n=9), Influenza A virus (n=3), Epstein-Barr virus (n=1), Toscana virus (n=1)).

**Controls:** Individuals who had a lumbar puncture performed because LNB was suspected, but where white-blood-cell (WBC) count in the CSF was within normal reference ranges (CSF-WBC: 0-5 x $10^6$/L) and the *B. burgdorferi* s.l. specific intrathecal antibody index was negative.

**Post-treatment-Lyme-disease-syndrome:** Patients with persisting symptoms of Lyme borreliosis >6 months after diagnosis and treatment.

***Erythema migrans*:** Patients with a skin rash clinically diagnosed with *erythema migrans* at the Unit for Tick- Borne Infections, Copenhagen University Hospital, Rigshospitalet.

***Acrodermatitis chronica atrophicans*:** Patients clinically diagnosed with *acrodermatitis chronica atrophicans* skin rash and positive *B. burgdorferi* s.l.IgG in blood at the Unit for Tick Borne Infections, Copenhagen University Hospital, Rigshospitalet.

### Proteomic analysis

CSF and plasma samples were thawed and 100 µl were transferred to 96-well plates for analysis. The sample preparation was optimized based on the previously published methods described in[44,45]. Briefly, 20 µl CSF was were first denatured with 30 µl PreOmics Lysis buffer, while 5 µl plasma was denatured

with 45 µl PreOmics Lysis buffer[46]. Both sample types were subsequently digested using LysC/trypsin enzyme mix. The resulting peptides were purified using two-gauge SDB-RPS StageTips, and the eluate analysed on Evosep One (Evosep Biosystem, Denmark) liquid chromatography system, coupled online to an Orbitrap Exploris 480 mass spectrometer. Data acquisition was performed in data-independent analysis (DIA), using 60 samples per day (SPD) gradient and 8 cm Pepsep column.

### Data processing

Initial data processing of the mass spectrometry raw files were performed with DIA-NN version 1.9 in a data independent search[47].

The DIA-NN data underwent further processing using the Clinical Knowledge Graph (CKG), alphapeptstats and Jupyter Notebook[48,49]. Initially, a stringent filter for missing data was applied: 1) samples with low protein count, defined by a value below 1.5IQR from the 25$^{th}$ quantile of the combined distribution, were excluded, and 2) proteins with a missingness of more than 40% across samples were excluded. Data was log2 transformed. The remaining missing values were imputed with a variational autoencoder using the PIMMS software[50]. Assessment of sample quality was conducted as previously described[50]. Batch correction was executed using combat to overcome potential plate-specific bias on subsequent analyses[51].

### Data analysis

Proteins exhibiting significantly different levels between the cohorts were identified by unpaired t-tests. Multiple hypothesis correction was applied with the Benjamini-Hochberg method, with adjusted P-values < 0.05 deemed statistically significant. P-values and protein abundances were visualized in Volcano plots, with -log10(corrected p-value) and log2 of protein fold change between groups. Venn diagrams were used to visualize overlaps in significant proteins between statistical comparisons. Heatmaps in combination with Sankey plots of gene ontology terms were used to highlight protein changes of the significant proteins together with their biological processes[52]. In the heatmaps, protein abundances were z-scored. Gene ontology terms were retrieved for each protein through UNIPROT. Gene ontology terms with frequency of less than 5% across proteins were not visualized.

### Model development

Z-scored data from the development cohorts were analysed using supervised machine learning to explore the potential for a diagnostic signature for LNB. Significant proteins identified by t-test analysis on the development cohorts were used as input features in the training data. Our data sets, comprising both samples from patients diagnosed with LNB and viral meningitis or control samples, was partitioned into a training set for model development and a test set for model validation using a 5-fold or 3-fold cross-validation (CV) approach for CSF and plasma respectively. The number of CV-folds was decided based on sample size with a minimum of 30 samples in each set. Both sets maintained an equal ratio of positive and negative cases (stratified k-fold cross-validation). The classification target used for analysis

was the diagnosis of LNB and viral meningitis or LNB and control (yes/no). An appropriate ML model was determined by testing the performance of twelve different algorithms based on the area under the curve (AUC) and Matthews Correlation Coefficient (MCC) in the development test set cross-validations. For each algorithm, optimal hyperparameters and features were selected during cross-validation. Feature importance were highlighted for the top 10 most predictive features with SHAP (SHapley Additive exPlanations) values[53].

### Model validation

The models trained on the development cohorts were subsequently applied to the validation cohorts and the performance was assessed by AUC and MCC. ROC curves and confusion matrices were used to visualize the performance of the classifiers.

## Figures

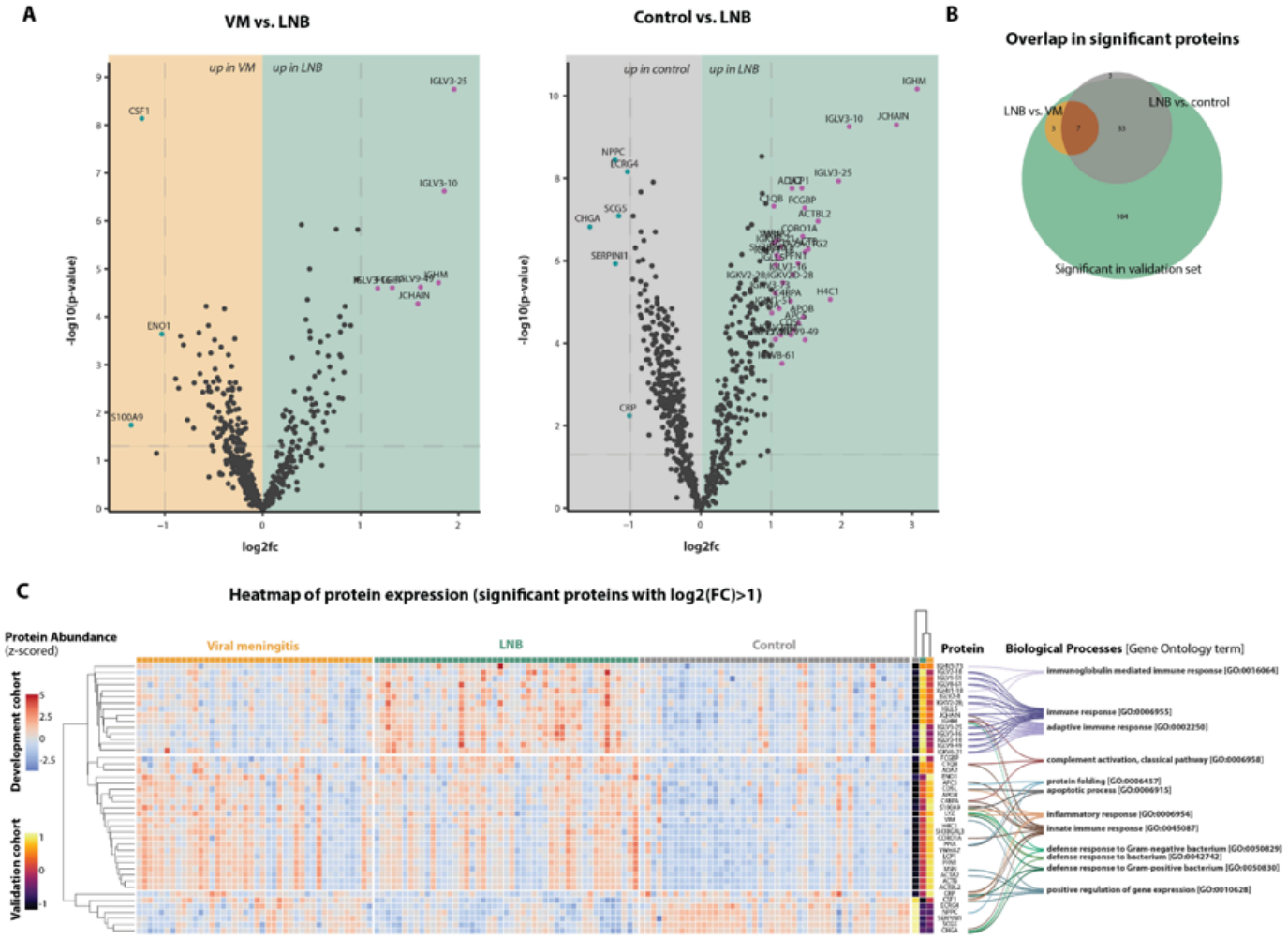
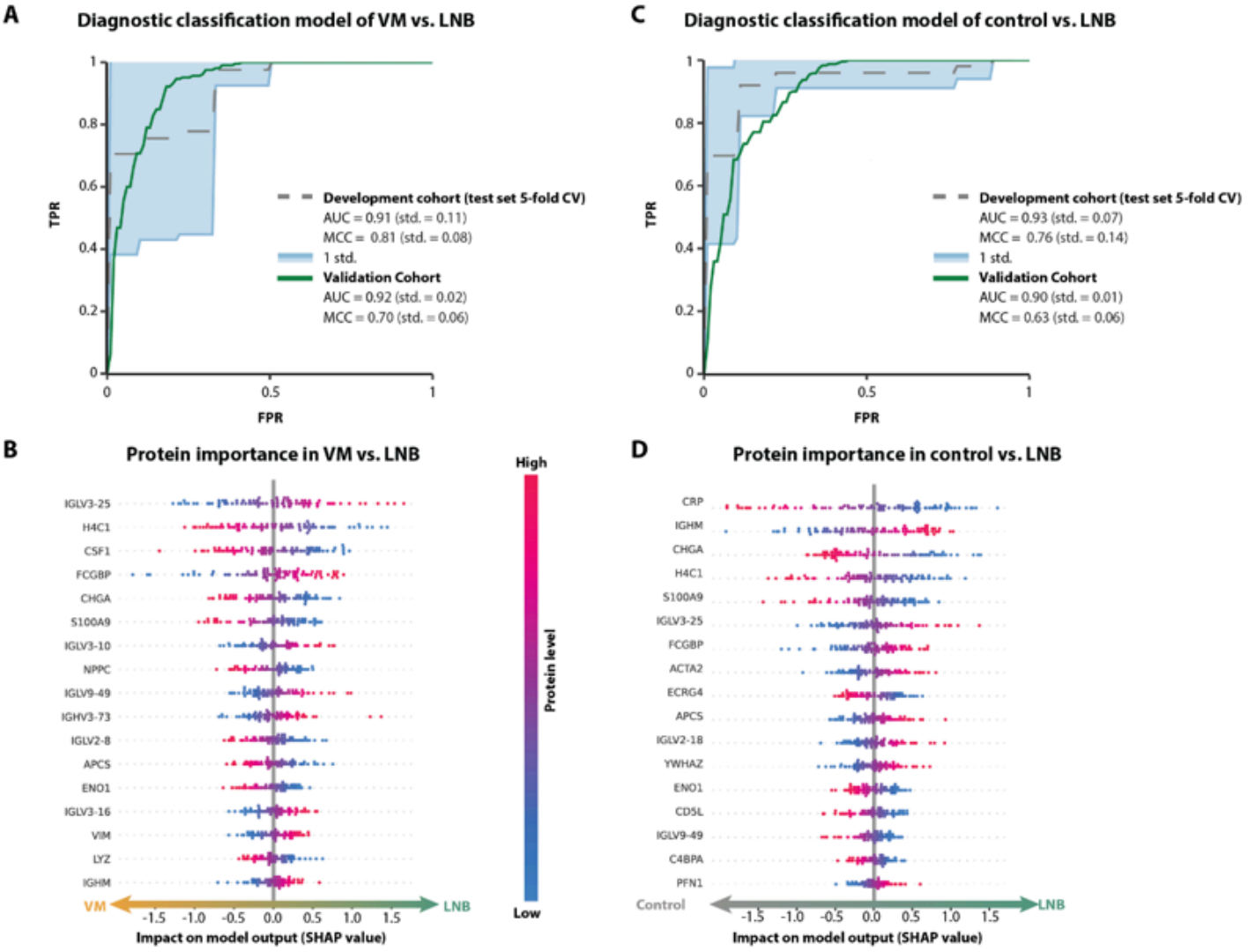



**Figure 1**

**A.** Volcano plots highlighting protein differences between Lyme neuroborreliosis (LNB) vs. viral meningitis (VM) and LNB vs. Controls in the cerebrospinal fluid development cohort. Each point represents a protein, the colour purple represents upregulated proteins with a log2 fold change (FC) larger than 1, the colour blue represents downregulated proteins with a log2 fold smaller larger than -1 and the colour grey represents proteins with a log2 fold change between -1 and 1 or non-significant proteins. X-axis depicts log2 fold change and y-axis depicts the log 10 adjusted p-value. Horizontal dashed lines represent the significance threshold after multiple hypothesis correction at p=0.05. Vertical dashed lines represent large log2 fold changes: above 1 or below -1. **B** Overlap in significant proteins from the two volcano plots with log2 fold changes lower than -1 or larger than 1. **C** Heatmap of the significant proteins for each sample in the development cohort and the corresponding mean from the validation cohort. The Gene Ontology-term associated with the protein is depicted on the right.




## Figure 2

**A** AUC for the final machine-learning model (Logistic Regression) developed to discriminate between LNB and viral meningitis (VM) cases in cerebrospinal fluid samples. Prediction performance is presented with Area Under ROC Curve (AUC) and Matthews Correlation Coefficient (MCC) for each cohort. Standard

deviation (std.) from the 5-fold cross validation (CV) is shown as semi-transparent error intervals for the test set. **B** Feature importance of the LR model for the top 10 most predictive features depicted with SHAP values. **C** AUC for the final ML-model (Support Vector Classifier (SVC)) developed to discriminate between LNB and controls. Prediction performance is presented with AUC and MCC for each cohort. Standard deviation from the 5-fold cross validation is shown as semi-transparent error intervals for the test set. **D** Feature importance of the SVC model for the top 10 most predictive features depicted with SHAP values. Abbreviations: TPR; True Positive Rate, FPR; False Positive Rate.

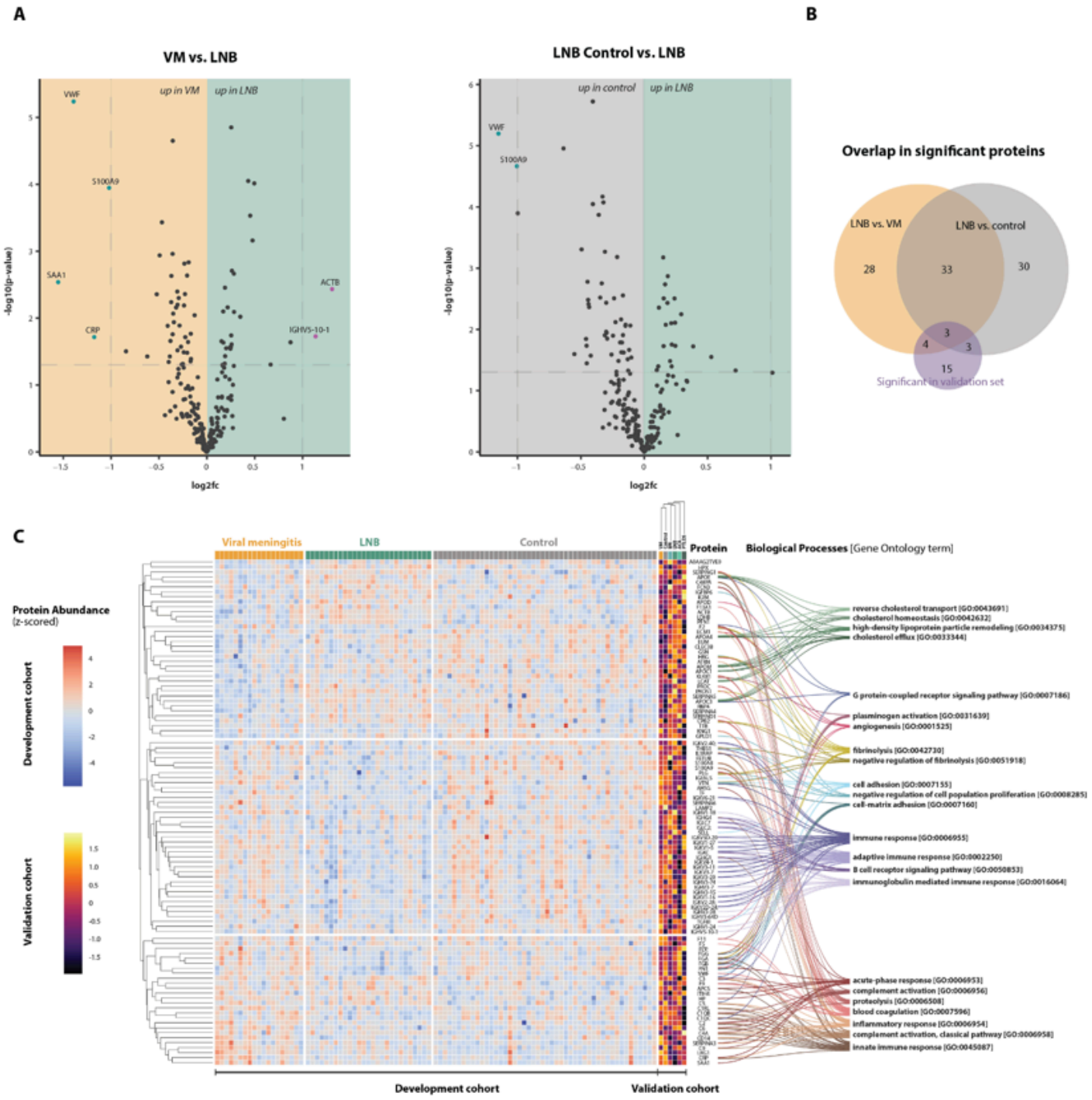
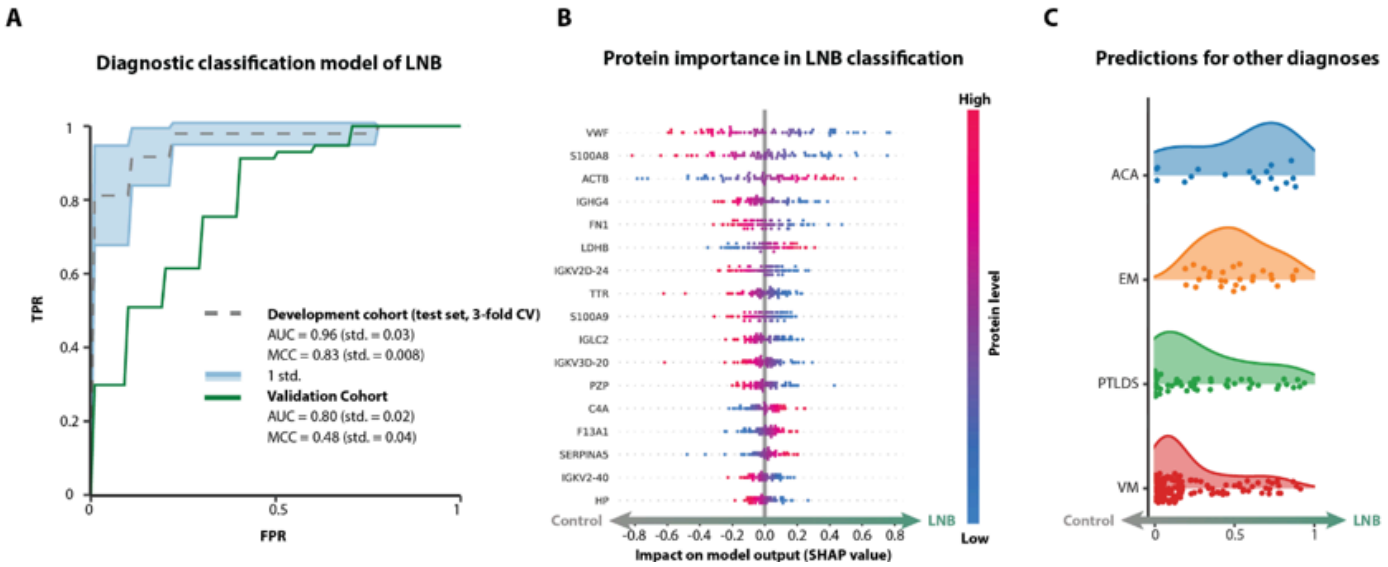


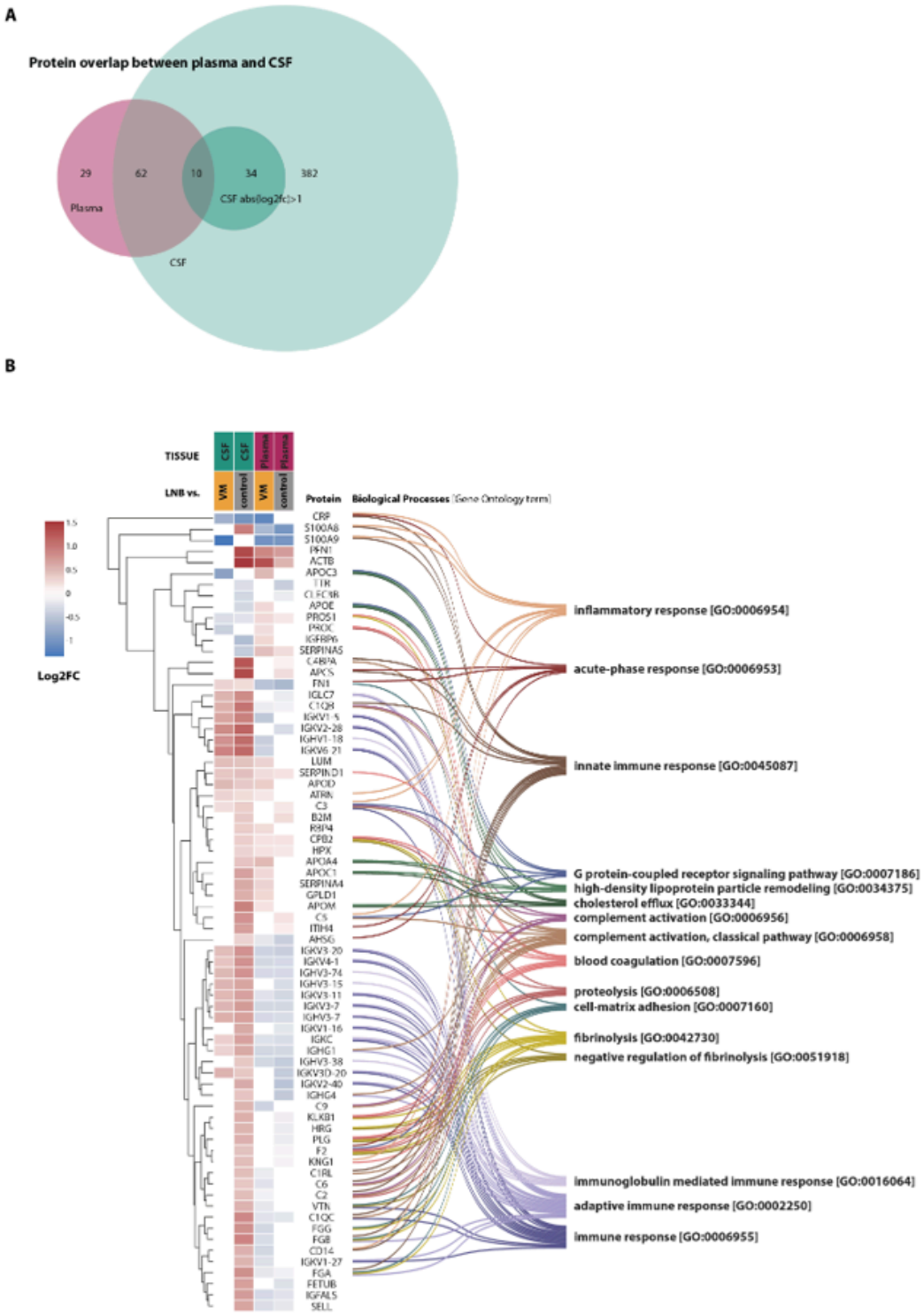

**Figure 3**

**A** Volcano plots highlighting protein differences between Lyme neuroborreliosis (LNB) vs. viral meningitis (VM) and LNB vs. Controls in the development cohort in plasma samples. Each point represents a protein, the colour purple represents upregulated proteins with a log2 fold change larger than 1, the colour blue represents downregulated proteins with a log2 fold smaller larger than -1 and the colour grey represents proteins with a log2 fold change between -1 and 1 or non-significant proteins. X-axis depicts log2 fold change and y-axis depicts the log10 adjusted p-value. Horizontal dashed lines represent the significance threshold after multiple hypothesis correction at p=0.05. Vertical dashed lines represent large log2 fold changes: above 1 or below -1. **B** Overlap in significant proteins from the two volcano plots with log2 fold changes lower than -1 or larger than 1. **C** Heatmap of the significant proteins for each plasma sample in the development cohort and the corresponding mean from the validation cohort. The Gene Ontology-term associated with the protein is depicted on the right.




## Figure 4

**A** AUROC for the final machine-learning (ML) model, Support Vector Classifier (SVC), developed to discriminate between Lyme neuroborreliosis (LNB) and controls in plasma samples. Prediction performance is presented with Area Under ROC Curve (AUC) and Matthews Correlation Coefficient (MCC) for each cohort. Standard deviation (Std.) from the 3-fold cross validation (CV) is shown as semi-transparent error intervals for the test set. **B** Feature importance of the SVC model for the top 10 most predictive proteins depicted with SHAP values. **C** Distributions of predicted probabilities from the LNB classification model on other diagnostic groups. Abbreviations: TPR; True Positive Rate, FPR; False Positive Rate, ACA; Acrodermatitis chronica atrophicans, EM; Erythema migrans, PTLDS; post-treatment-Lyme-disease-syndrome, VM; Viral meningitis.




## Figure 5

Comparison of cerebrospinal fluid (CSF) and plasma proteins. **A**Venn diagram highlighting overlap between significant proteins in the comparisons in the different tissues. **B** Heatmap of the log2 fold changes between LNB and VM or controls of the 72 overlapping proteins. Empty fields represent non-significant findings for that comparison. Gene ontology terms for the proteins are summarized in the Sankey plot on the right.

## Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- SupplementaryMaterial.docx